

CPU vs Memory: A Beginner's Cloud Resource Guide

Understanding the difference between CPU and Memory stops you from wasting money on oversized cloud servers. It helps you quickly debug why your application crashed or slowed down during peak hours.

What you'll learn:

- 1 Checking Free Memory with 'free'
- 2 Tracking Active Processes with 'htop'
- 3 Inspecting Docker Containers with 'docker stats'
- 4 Finding the Top RAM Hog with 'ps'

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Checking Free Memory with 'free'

The 'free -h' command displays the total amount of free and used physical memory and swap space in the system in a human-readable format. This is your first line of defense when an application crashes due to lack of memory. It helps you see if your system is resorting to slow swap space on the disk.

```
$free -h
```

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Tracking Active Processes with 'htop'

'htop' is an interactive process viewer that gives you a color-coded, real-time look at your CPU cores and memory usage. It lets you easily sort processes by resource consumption and even kill misbehaving processes directly from the interface. It's much easier to read than the classic 'top' command.

```
$htop
```

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Inspecting Docker Containers with 'docker stats'

When running containerized applications, you need to know how much resources each container is consuming. The 'docker stats' command provides a live stream of CPU, memory, network, and block I/O usage for all running containers. This helps you set accurate resource limits in your Docker Compose files.

```
$docker stats --format "table {{.Name}}\t{{.CPUPerc}}\t{{.MemUsage}}"
```

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Finding the Top RAM Hog with 'ps'

Sometimes a single hidden background process is eating up all your server's memory. By combining 'ps' with sorting flags and 'head', you can quickly print the top 5 memory-consuming processes on your Linux system. This is perfect for quick troubleshooting during a live incident.

```
$ps aux --sort=-%mem | head -n 6
```


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Key Takeaways

- + CPU represents processing speed (the brain); Memory/RAM represents active workspace (the desk).
- + High CPU causes slowness; low Memory causes sudden application crashes (OOM kills).
- + Swap space is disk space used as emergency memory, but it is extremely slow.
- + Always set resource limits on containers to prevent one container from crashing the whole host.
- + Use 'htop' for interactive debugging and 'ps' for quick scripting and sorting.

Watch Out For

- ! Adding more CPU to fix an Out-Of-Memory (OOM) crash.
- ! Ignoring 'swap' usage, which secretly slows down your entire application.
- ! Forgetting to set memory limits on Docker containers, allowing one container to hog the entire host.
- ! Assuming cloud auto-scaling will instantly save you from bad code that leaks memory.

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